

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 4: PANS-OPS Instrument Procedures & Circling

Syllabus Source: CAE Oxford (pp. 117-156) | Pass Target: $\geq 90\%$ AviationExam

1. PANS-OPS INSTRUMENT DEPARTURE CRITERIA

ICAO Doc 8168 (PANS-OPS) establishes standard criteria for protected airspace design:

- Design Climb Gradient (PDG): Standard departure climb gradient is 3.3% (200 ft/NM), comprising 2.5% net obstacle clearance + 0.8% safety buffer.
- Minimum Turn Altitude: No turn may be initiated on departure below 120 METERS (400 FT) ABOVE AERODROME ELEVATION!

2. VISUAL MANOEUVRING (CIRCLING APPROACH)

A visual circling approach is used when landing runway does not align with the instrument approach (alignment $> 30^\circ$ off runway centerline, or descent gradient exceeds 6.5%):

- Protected Circling Radii: Cat A = 1.68 NM; Cat B = 2.66 NM; Cat C = 4.20 NM; Cat D = 5.28 NM from all runway thresholds.
- Action on Loss of Visual Reference: IF VISUAL CONTACT WITH THE RUNWAY IS LOST DURING CIRCLING: Initiate an IMMEDIATE CLIMBING TURN TOWARDS THE LANDING RUNWAY, establish overhead the aerodrome, and follow the missed approach procedure of the INSTRUMENT RUNWAY originally flown!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Circling Missed Approach Procedure

Golden Rule: If visual reference is lost at any point during visual circling to land: Step 1: Apply immediate TOGA thrust and initiate positive climb. Step 2: Turn TOWARDS the landing runway centerline to stay within protected obstacle clearance area. Step 3: Establish the aircraft on the published missed approach procedure of the initial instrument approach! Fatal Trap: Never turn away from the runway into unprotected airspace during a circling go-around.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under PANS-OPS design rules, what is the MINIMUM altitude above aerodrome level before an aircraft may

- [A]: 200 ft
- [B]: 120 METERS (400 FEET) AGL
- [C]: 1,000 ft
- [D]: 500 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS prohibits turns below 120 m (400 ft) above aerodrome elevation to guarantee obstacle clearance during straight acceleration.

Q2: If visual reference is LOST while conducting a visual circling manoeuvre to land, what is the mandatory pil

- [A]: Descend 500 ft to find the ground
- [B]: Initiate an IMMEDIATE CLIMBING TURN TOWARDS THE LANDING RUNWAY, climb overhead, and fly the missed approach of the instrument approach runway
- [C]: Turn 90° away from airport
- [D]: Land immediately

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Turning towards the runway keeps the aircraft within the protected circling obstacle area while climbing to safety in the instrument missed approach.

Q3: What is the radius of the protected obstacle clearance area for a CATEGORY C aircraft conducting a circling approach?

- [A]: 1.68 NM
- [B]: 4.20 NAUTICAL MILES from the runway threshold
- [C]: 2.66 NM
- [D]: 5.28 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS assigns Cat C aircraft a circling area radius of 4.20 NM based on a maximum circling speed of 180 kt.